

FSK116: Tutorial Problem Numbers / Tutoriaal Probleem Nommers

2017

12 Week: 22 May 2017 / 22 Mei 2017

All the indicated numbers refer to the “Problems” section in Serway; e.g. P1.9 is problem 9 from chapter 1.

Alle nommers verwys na die “Problems” afdeling in Serway; bv. P1.9 is probleem 9 uit hoofstuk 1.

P15.5 (a) 1.50 Hz (b) 0.667 s (c) 4.00 m (d) π rad (e) 2.83 m

P15.13 (a) -2.34 m (b) -1.30 m/s (c) -0.0763 m (d) +0.315 m/s

P15.14 (a) Motion is periodic. (b) 1.81 s (c) Motion is not simple harmonic motion.

P15.29 (a) $8/9E$ (b) $1/9E$ (c) $x = \pm\sqrt{2/3 \cdot A}$ (d) No

P15.36 $\omega = \sqrt{k/m} = \sqrt{g/R}$

P15.43 (a) 3.65 s (b) 6.41 s (c) 4.24 s

P15.64 (a) 2.00 cm (b) 4.00 s (c) $\pi/2$ rad/s (d) π cm/s (e) 4.93 cm/s^2
(f) $x = 2.00 \sin(\frac{\pi}{2}t)$

P16.2 (a) graph (b) graph (c) Amplitude and wavelength are the same.
(d) 2.40 m to the right.

P16.11 (a) $\omega = 31.4 \text{ rad/s}$ (b) $k = 1.57 \text{ rad/m}$ (c) $y = (0.120 \text{ m}) \sin[(1.57 \text{ rad/m})x - (31.4 \text{ rad/s})t]$ (d) 3.77 m/s (e) 118 m/s^2

P16.12 $\pm 6.67 \text{ cm}$

P16.13 (a) 0.500 Hz (b) 3.14 rad/s (c) 3.14 rad/m (d) $y = (0.100 \text{ m}) \sin[(\pi \text{ rad/m})x - (\pi \text{ rad/s})t]$ (e) $y = (0.100 \text{ m}) \sin[(\pi \text{ rad/s})t]$ (f) $y = (0.100 \text{ m}) \sin[(4.71 - (\pi \text{ rad/s})t)]$ (g) 0.314 m/s

P16.18 (a) graph (b) 18.0 rad/m (c) 0.0833 s (d) 75.4 rad/s (e) 4.20 m/s
(f) $y = (0.200 \text{ m}) \sin[(18.0 \text{ rad/m})x + (75.4 \text{ rad/s})t + \phi]$